NGUYEN DUY TRANG

MOBILE ENGINEER

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PROFESSIONAL SUMMARY

OVERVIEW

- **Core Mobile Engineering:** 5+ years of experience building and delivering production mobile applications across Flutter, Swift (iOS), and Kotlin (Android), with deep native platform integrations.
- **Adaptability & Problem Solving:** Demonstrated versatility across Social OTT, Fintech, POS checkout software, self-order Kiosks, and on-device AI integration. Pragmatically blends Native (iOS/Android) and Flutter capabilities to maximize performance, delivery speed, and business goals.
- **Team Collaboration & Ownership:** Proactive and highly collaborative across Product, UI/UX, Backend, and QA teams. Actively contributes practical technical ideas to resolve blockers and supports teammates to ensure smooth, on-time delivery.

KEY HIGHLIGHTS & ACHIEVEMENTS

METRICS

- **Measured Performance Optimization:** Reduced self-order Kiosk checkout time by 30%; optimized camera buffer streaming to achieve sub-2.5s AI inference latency with locked 60 FPS rendering (0 memory leaks).
- **Offline-First — Local Database:** Embedded a high-performance Rust core into Flutter via Dart FFI to directly manage the Local Database (SQLite) and execute transactional business logic, guaranteeing zero-latency response and 100% offline persistence.
- **Automated Unit Testing & Self-Testing:** Champions a rigorous self-testing culture: personally authored and maintained 277 automated unit and widget tests (100% pass) covering business logic and UI states, integrated into automated CI/CD pipelines to guarantee zero regressions.
- **1M+ Production Scale & Reliability:** Proven track record operating large-scale multi-app mobile systems reaching 1M+ downloads across iOS and Android, consistently maintaining a 99.5%+ crash-free session rate in production.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Platforms:	Flutter & Dart, Swift (iOS), Kotlin (Android), Rust, TypeScript.
Architecture & State:	Clean Architecture, Riverpod, BLoC, MVVM, VIPER, Modularization.
Networking & Data:	Offline-First, SQLite, Realm, Secure Storage, REST APIs (Dio), WebSocket, Firebase, Dart FFI.
Performance & Advanced:	Flutter DevTools, Render Tree Optimization, AutoRoute, Gemini AI, GenUI (A2UI).
Testing & CI/CD:	Unit & Widget Testing (277 Tests), GitHub Actions, GitLab CI/CD, Fastlane.
Delivery & Monitoring:	App Store Connect, Google Play Console, Firebase App Distribution, Sentry, Crashlytics, GA4.

FEATURED FLAGSHIP PROJECT

PERSONAL ARCHITECTURE

ASTROBITE • **Mobile Engineer & System Architect** (AI Calorie Tracker & Food Scanner) 2026 – Present

Tech Stack: Flutter, Dart, Clean Architecture, Riverpod 2.x, AutoRoute, Gemini AI, Dio, SQLite, Fastlane, CI/CD.

- Architected AstroBite (v3.0.0) based on **Clean Architecture / Modular Package-based** structure with Riverpod 2.x code-gen across 12 decoupled business modules.
- Packaged proprietary **2D/3D Design System & Base Components** natively in Flutter (Custom Widgets, Theme Engine, Responsive Breakpoints), locking consistent **60 FPS**.
- Constructed **Base Network Client & Local Persistence**: Dio Interceptors (token refresh, retry policy), WebSocket & Firestore stream real-time, Hive/SQLite offline caching.
- Optimized **Render Tree & Memory via Flutter DevTools**: isolated painting via `RepaintBoundary`, `const` constructors, camera buffer disposal yielding sub-2.5s AI latency and **0 memory leaks**.
- Engineered deep-linking and web navigation architecture using **AutoRoute**: supporting nested routes, route guards, URL query params, CanvasKit/WASM rendering, and asset caching.
- Established a suite of **277 automated unit & widget tests (100% pass)**; standardized CI/CD pipelines (Web/Android/iOS) via Fastlane, GitHub Actions & GitLab CI/CD.

COMMERCIAL WORK EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY • Mobile Developer (Flutter, POS & Kiosk)

08/2025 – 08/2026

Tech Stack: Flutter, Dart, Clean Architecture, BLoC, Dio, SQLite/Drift, Platform Channels (Kotlin), Fastlane, CI/CD.

- Architected modular **Clean Architecture / MVVM** applying **BLoC (flutter_bloc)** with Unidirectional Data Flow for multi-vertical POS & Kiosk checkouts (F&B, Retail, Spa), slashing transaction time by **30%**.
- Packaged **Base UI Library & Responsive Layout** dynamically adapting across screen form-factors (Mobile, Tablet, Kiosk/Web) with reusable Custom Widgets and dynamic Theme Engine.
- Built **Base Network Client with Dio**: configured Interceptors for automatic bearer token injection, 401 token refresh cycle, exponential backoff retry policy, and offline cache.
- Designed robust **Offline-first** data pipeline (BLoC, SQLite/Drift), ensuring 100% order persistence during network disconnects and auto-sync upon reconnection.
- Profiled Render Tree and eliminated frame drops/jank using **Flutter DevTools** (Profiler, Memory), preventing RAM leaks while driving ESC/POS thermal printers and barcode scanners via Kotlin Platform Channels.
- Established **Coding conventions**, strict lint rules (`analysis_options.yaml`), led PR code reviews, and **mentored Junior developers** across the engineering team.
- Configured automated CI/CD build & release pipelines (Android APK/AAB, Web bundle) via **Fastlane & GitHub Actions / GitLab CI/CD**.

HAHALOLO SOCIAL TRAVEL NETWORK •

05/2020 – 06/2025 (5 years)

Mobile & Frontend Developer (iOS & Flutter)

Tech Stack: Flutter, iOS Swift, VIPER, MVVM, Riverpod, BLoC, WebSocket, SQLite, CallKit, Crashlytics, CI/CD.

- **[Hahalolo — Travel Social Network (1M+ downloads)]**: Developed Tour, Flight, Hotel, Community modules; built Native iOS (Swift, VIPER) and integrated **Flutter via Add-to-App**; handled Responsive UI, Deep Linking, Crashlytics <0.5% crash rate. Mobile team: 20.
- **[Halome — OTT Realtime Messaging]**: Engineered real-time chat app from scratch using Native iOS Swift (MVVM); integrated **WebSocket real-time**, SQLite offline storage, RxSwift, CallKit VoIP; optimized for locked 60 FPS list scrolling. Team: 5.
- **[Haloki & Hahalolo Invest — Fintech]**: Engineered and optimized international remittance app (192 countries) and stock trading platform in Flutter; implemented secure KYC verification, local encrypted persistence (SQLite/Keychain), and smooth real-time financial charting.
- **[Aquannets — Outsource for Galactic Holdings]**: Developed aquaculture farm management app in Flutter applying **Clean Architecture + Riverpod**, SQLite offline-first at ponds. Team: 2.
- **[SocTrip — QC & Testing Lead]**: Led end-to-end (E2E) testing lifecycle, designed comprehensive test matrices, and provided quality gate sign-off for App Store & Google Play releases.

EDUCATION

ACADEMIC

FPT POLYTECHNIC

LANGUAGES

PROFICIENCY

English: Reads and understands technical documentation